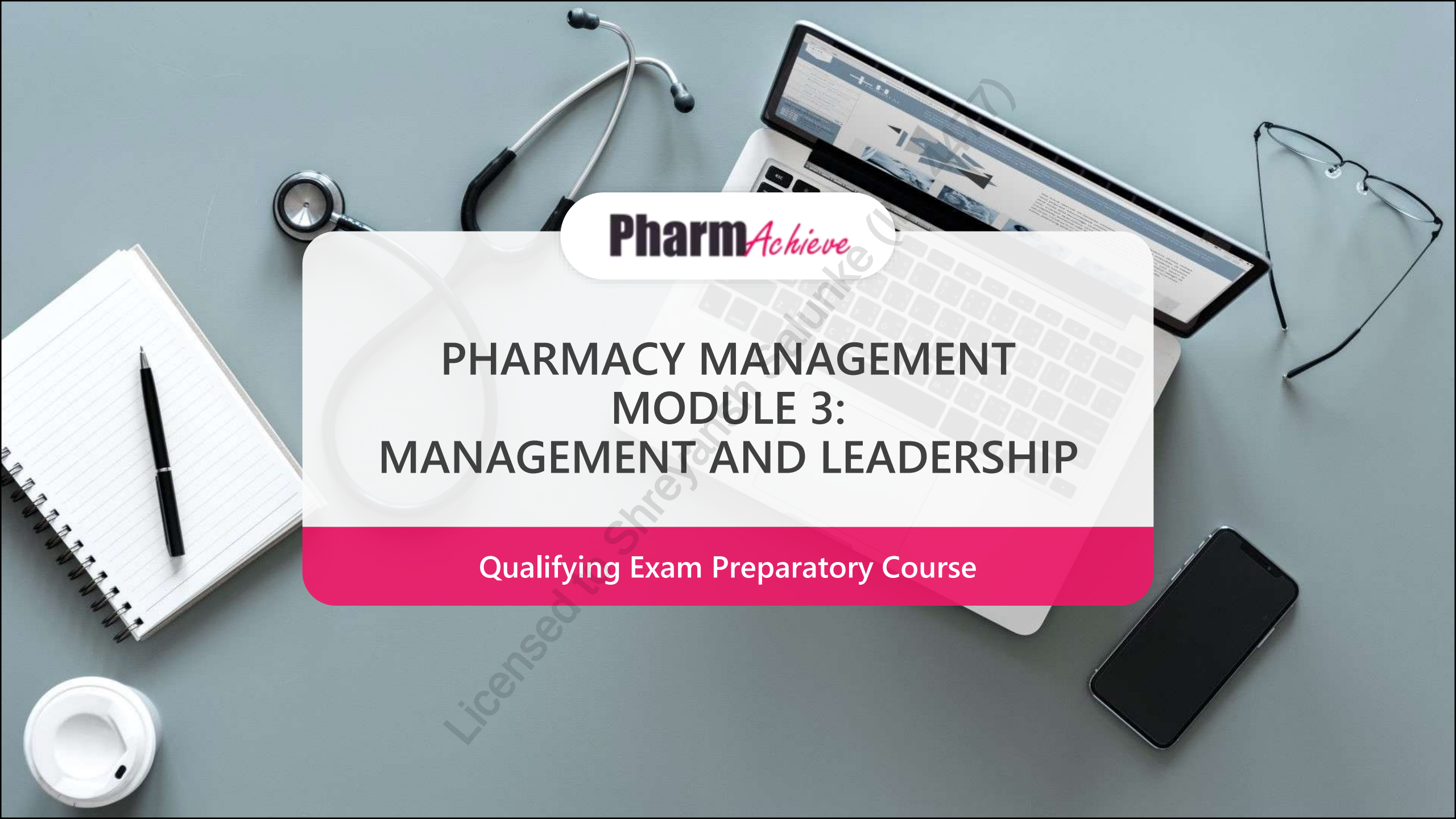




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PHARMACY MANAGEMENT MODULE 3: MANAGEMENT AND LEADERSHIP

Qualifying Exam Preparatory Course

COVERED IN THIS PRESENTATION...

Ethical, Legal and
Professional
Responsibilities

Patient Care

Product
Distribution

Practice Setting

Health Promotion

Knowledge and
Research
Application

Communication
and Education

Intra and Inter-
Professional
Collaboration

Quality and Safety

LEGEND

ACE inhibitor Angiotensin-Converting Enzyme Inhibitor

ADR Adverse Drug Reaction

API Active Pharmaceutical Ingredient

ARB Angiotensin Receptor Blocker

COGS Cost of Goods Sold

DM Designated Manager

H₂RA H₂ Receptor Antagonist

ITR Inventory Turnover Rate

NPN Natural Product Number

PPI Proton Pump Inhibitor

TABLE OF CONTENTS

1 PEST (Political, Economic, Social, and Technological) Analysis and Strategic Planning

- Macro-environmental Analysis
- Strategic Planning

2 Financial Statements

3 Designated Manager

4 Continuous Quality Improvement

5 Administrative Considerations

6 Common Workplace Departments

PEST (Political, Economic, Social, And Technological) ANALYSIS AND STRATEGIC PLANNING

Qualifying Exam Preparatory Course



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MACRO-ENVIRONMENTAL ANALYSIS

Pharmacy organizations are affected by external and internal environments:

Internal

- Controllable by the organization (human resources, financial resources (capital assets), physical resources, technological resources, operating systems, policies and products and services produced

External

- Not generally controllable by the organizations (legislation, economy, social patterns and technology)

To identify the impact of changes in the macro-environment, we conduct a **PEST analysis**:

P Political

E Economic

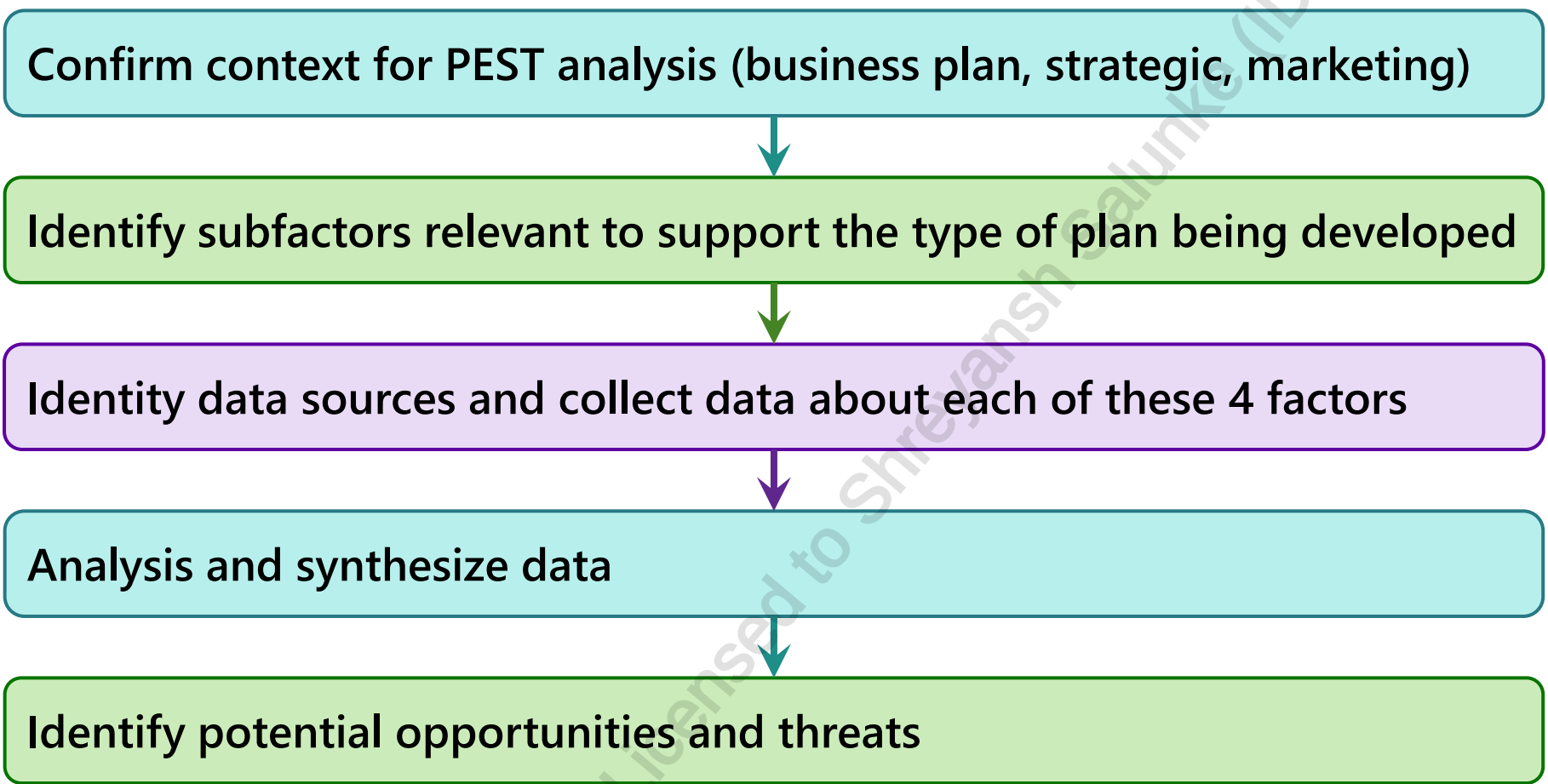
S Social

T Technological

MACRO-ENVIRONMENTAL ANALYSIS

To conduct a PEST analysis:

Confirm context for PEST analysis (business plan, strategic, marketing)



```
graph TD; A[Confirm context for PEST analysis (business plan, strategic, marketing)] --> B[Identify subfactors relevant to support the type of plan being developed]; B --> C[Identify data sources and collect data about each of these 4 factors]; C --> D[Analysis and synthesize data]; D --> E[Identify potential opportunities and threats];
```

The flowchart consists of five rectangular boxes arranged vertically, connected by downward-pointing arrows. The boxes are colored as follows: light blue, light green, light purple, light blue, and light green. The text inside each box is black and bold. A faint watermark 'Licensed to Shreyansh Salunkhe (ID: 4477)' is visible diagonally across the center of the flowchart.

Identify subfactors relevant to support the type of plan being developed

Identify data sources and collect data about each of these 4 factors

Analysis and synthesize data

Identify potential opportunities and threats

MACRO-ENVIRONMENTAL ANALYSIS

SAMPLE PEST ANALYSIS

P	Political	federal, provincial, laws, regulations and legislation • Example: <i>Individual provincial, territorial drug plan policies dictate eligibility, cost-sharing mechanisms, drug pricing, professional fees, markups, co-ordination of benefits</i>
E	Economic	economic growth, interest and inflation rates • Example: <i>Drug patents are expiring, which results in increased availability of generic drugs → savings to public and private drug plan expenditures (also biologics and biosimilars)</i>
S	Social	population growth rate, age distribution, work attitudes and cultural factors, patients choosing OTC products and self-care over prescriptions
T	Technological	health care technology is the application of organized knowledge and skills in the form of devices, medicines, vaccines, procedures, and systems developed to solve health problems and improve quality of lives • Example: <i>POS point of sale system, e-health prescribing</i>

MACRO-ENVIRONMENTAL ANALYSIS

SAMPLE PEST ANALYSIS: Identifying potential opportunities and threats

Analysis	Opportunities	Threats
Political	Expanded scope of practise, more professional billing	Variable and/or limited pharmacist uptake of the expanded scope of practise
Economic	Pharmacy advocacy to seek adjustments to current professional fees and services	Public and private payers implementing cost constraints to reimbursement programs
Social	Continued demand for pharmacist consultations, minor ailments OTC recommendations	Other health care professionals will expand their areas such as drug distribution, immunization, prescribing, medication management
Technological	Drug information systems such as e-prescribing	Poor rates of pharmacy uptake of technology

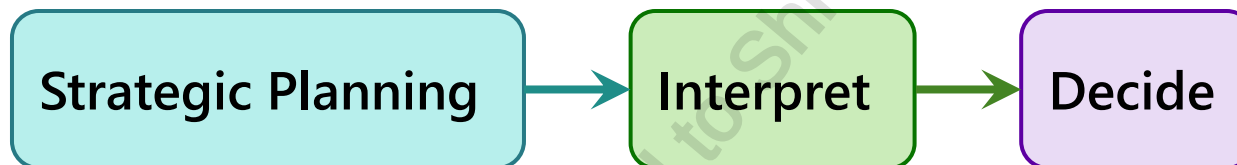
FOUNDATIONS FOR A SUCCESSFUL STRATEGIC PLAN

1. Strategic planning

"Charting a course of where to go, when to get there and how to get there (achieving goals)"

Environmental scan and market analysis:

- Assessment of the physical location of the pharmacy (street access and parking)
- Complete list of competitors (including key services, products and pricing offered by each)
- Demographics of the area (such as age, gender, income levels)



- **Market niche:** selling a product that targets a specific/unique population
- This terminology is commonly used interchangeably with **market segment**, but the difference is that the target population in market segment is larger than market niche

FOUNDATIONS FOR A SUCCESSFUL STRATEGIC PLAN

2. Interpret

Properly interpreting and analyzing all the information collected during the scanning process is critical to making good decisions that will form the basis of the strategic plan

One common method to analyze data collected is to perform a SWOT analysis:

S

Strengths

Characteristics of your business that may be viewed as unique or different and are likely to be seen as giving one an advantage over competitors

W

Weaknesses

Characteristics of a business that may place one at a disadvantage relative to competitors

O

Opportunities

Areas that if implemented, could provide an advantage over others, thus can become strengths

T

Threats

Elements identified in an environmental scan and market analysis that can cause difficulty for a business and threaten its viability and success

FOUNDATIONS FOR A SUCCESSFUL STRATEGIC PLAN

3. Decide

Once the information is collected in the environmental scan and the market analysis has been categorized, the next step is to decide how to use that information in creating a finalized plan

Setting SMART goals:

Specific

- Who
- What
- Where
- When
- Why

Measurable

- Ensuring criteria that can be measured toward the achievement of the goal
- When progress is measured, it is easier to track

Achievable

- Set goals that can truly be achieved
- Focus on being able to use the skills, abilities and resources available

Realistic

- Set goals that one is motivated to achieve
- Feeling motivated and constantly measuring progress usually means the goals are realistic

Timely

- Set goals that can be achieved within a reasonable time frame and that will have a positive effect on the business

FINANCIAL STATEMENTS

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FINANCIAL STATEMENTS

- Provides a picture of your business' financial status over a stated period
 - Usually a fiscal quarter (3 months) or a fiscal year (12 months)
- Should be used at **least on a yearly basis** to review progress of your business & assess how to improve
- The **three core financial statements** are:
 - Balance sheet
 - Income statement
 - Statement of cash flows

Who looks at financial statements?

- Stakeholders such as investors, creditors, pharmacy owners/operators – to assess management stewardship and the overall financial health of a pharmacy business

FINANCIAL STATEMENTS

Balance Sheet

- Snapshot of an organization's financial position at a specific point in time (a STATIC document)
- Key components: assets, liabilities, and shareholder equity
- There is no information about trends, how income was generated, types of expenses accrued, etc.

Income Statement

- A DYNAMIC document highlighting profit and loss over a specific time period
- Key components: revenues, cost of goods sold, gross profit, operating expense and net income (or net loss)
- Shows you a trend over time

Cash Flow Statement

- Focuses on the inflows and outflows of cash. How does the pharmacy generate and use cash
- Key components: operating activities, investing activities, and financing activities

FINANCIAL STATEMENTS

BALANCE SHEET

$$\text{Shareholders' Equity} = \text{Assets} - \text{Liabilities}$$

Shareholders Equity

- The owners' funds
- Common stocks → Ownership in the company
- Retained earnings → Profits reinvested into company
- Dividends → Payments to shareholders



Assets

- Things you own
- Current assets → Cash, accounts receivable, inventory
- Fixed assets → Real estate, equipment



Liabilities

- Money you owe others
- Current liabilities → Accounts payable to wholesalers, salaries payable to employees
- Fixed liabilities → Bonds payable

FINANCIAL STATEMENTS

BALANCE SHEET EXAMPLE

November 1	Year 0 (\$)	Year 1 (\$)
Shareholders' Equity		
Common stock	300,000	300,000
Retained earnings	0	200,000
Total shareholders' equity	300,000	500,000
Total liabilities and shareholders' equity	600,000	950,000

November 1	Year 0 (\$)	Year 1 (\$)
Assets		
Current assets		
Cash	50,000	100,000
Accounts receivable	0	200,000
Inventories	200,000	300,000
Noncurrent assets		
Building	300,000	300,000
Equipment	50,000	50,000
Total assets	600,000	950,000

Liabilities		
Current liabilities		
Accounts payable to wholesalers	200,000	170,000
Salaries payable to employees	0	30,000
Noncurrent liabilities		
Bonds payable	100,000	250,000
Total liabilities	300,000	450,000

FINANCIAL STATEMENTS

INCOME STATEMENT

$$\text{Net Income} = \text{Total Revenue} - \text{Total Expenses}$$

Net Income

- Overall amount of money remaining
- Positive → Profits
- Negative → Losses



Revenue

- Money coming in
- Sales → prescriptions, clinical services, OTC sales



Expenses

- Cost of business
- Cost of goods sold
- Administrative fees
- Income tax

FINANCIAL STATEMENTS

INCOME STATEMENT

Year 1 Example

Revenue	
Sales (prescription)	2,000,000
Sales (nonprescription)	500,000
Total revenue	2,500,000
Expenses	
Costs of goods sold	1,800,000
Administration expenses	350,000
Income tax expense	50,000
Total expenses	2,200,000
Net income	300,000

Money coming in

Costs of business

Positive = Profits

* Includes \$1,200,000 in total credit sales.

FINANCIAL STATEMENTS

CASH FLOW STATEMENT

How cash is distributed

Operating Activities

- Money used to keep the business running
- Includes revenues and expenses

Investing Activities

- Spending on new technologies/assets

Financing Activities

- Obtaining and distributing money
- E.g. issuing bonds, dividends

FINANCIAL STATEMENTS

STATEMENT OF CASH FLOWS

Year 1 Example

Operating	
Revenues providing cash	2,300,000
Expenses using cash	2,150,000
Cash flow from operating	150,000

Investing	
Acquisition of noncurrent assets	0
Sales of noncurrent assets	0
Cash from investing	0

Financing	
Issue of bond	0
Dividends	(100,000)
Net change in cash	(100,000)
Net change in cash	50,000
Cash (end of year 0)	50,000
Cash (end of year 1)	100,000

FINANCIAL RATIOS

Allows for better understanding of an organization's financial state
Compares the organization's financial state to the industry average (higher ratios = better)

Profitability ratios (ability to generate profit)

- **Gross profit margin** =
[(Revenue-COGS*)/Revenue] x 100%
- **Net profit margin** =
[Net income after taxes/Revenue] x 100%
 - Assesses a company's ability to generate profits
- **Return on assets** = net income/average total assets
 - Assesses the company's ability to generate profits using assets
- **Return on equity** = net income/average owner's equity
 - Assesses the company's ability to generate profits using shareholders' equity

Liquidity ratios (ability to address short-term finances)

- **Current ratio** = current assets/current liabilities
 - Shows the company's risk tolerance
 - Ability to pay bills and outstanding loans
- **Quick ratio** = (current assets-inventories-prepaid expenses)/current liabilities
 - Accurate for short-term relief

Turnover ratios (ability to efficiently use assets)

- **Inventory turnover ratio** = COGS/average inventory
 - Low, indicates inefficiency
 - High, shows susceptibility to drug shortages

FINANCIAL RATIOS

EXAMPLE

A new pharmacy owner is trying to calculate the gross profit margin for his sales using the following financial statement from his pharmacy.

Help him express his gross margin as a percentage using his income statement.

CASE STUDY

Revenue	
Sales (prescription)	2,000,000
Sales (nonprescription)	500,000
Total revenue	2,500,000
Expenses	
Costs of goods sold	1,800,000
Administration expenses	350,000
Income tax expense	50,000
Total expenses	2,200,000
Net income	300,000

* Includes \$1,200,000 in total credit sales

FINANCIAL RATIOS

EXAMPLE

CASE STUDY

RK is a pharmacy owner and is trying to calculate her inventory turnover rate for the year 2023.

Help RK calculate the turnover rate and interpret the results for her.

Is the turnover rate high or low?

RK Pharmacy inventory information for 2023

- COGS = \$400,000
- Beginning inventory = \$50,000
- Ending inventory = \$75,000

DESIGNATED MANAGER

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DESIGNATED MANAGER

Leadership and Management



The terms leadership and management are quite often used as though they were synonymous, although there are overlaps, leadership involves a different focus and requires a different set of competencies than management does

- *Pharmacy Management in Canada*

Manager	Leader
Administers	Innovates
Accepts the status quo	Challenges status quo
Has short range of view	Has long range of view and perspectives
Maintains	Develops
Focuses on systems and structure	Focuses on people
Relies on control	Inspires trust
Values stability	Values adaptability and change
Asks how and when	Asks what and why

Leadership involves envisioning what the organization will look like in the future, has long-term focus on doing the right things to ensure the organization remains vibrant and successful.

Management has a focus on doing things right and being efficient in executing the plans that leaders develop.

DESIGNATED MANAGER

EFFECTIVE LEADERSHIP SKILLS

Communicate concisely and effectively

Use emotional intelligence when dealing with difficult situations

Use active listening and watch for non-verbal clues

Conflict management

Motivate and inspire others for leadership roles

Compromise to achieve common goals

Make clear decisions based on evidence provided instead of emotions

Manage performance issues without demotivating staff

Prioritize tasks with proper planning

Promote team building within the workplace

Set goals and implement quality improvement processes to meet targets

Show enthusiasm and involve staff in decision making

Problem solve effectively to achieve goals

DESIGNATED MANAGER

What is the role of a designated manager?

- Manages the environment, which includes the premises of the pharmacy along with the equipment, systems, and staffing required to protect against and mitigate risks associated with the delivery of services
- Establish the culture to support pharmacy professionals to meet the standards of professional practice
- Ensures the standards of operation are being met:
 - Governance and legal compliance
 - Management and employee relations
 - Pharmacy premises and environment
 - Delivering services
 - Equipment and technology
 - Quality improvement and medication safety

DESIGNATED MANAGER

Operations management

Ensuring execution of the vision and mission of the organization (e.g. operations manager would be the pharmacy manager)

Involves many areas to achieve success such as:

- Management of staff and their assigned roles
- Developing processes to efficiently provide patient care
- Safety initiatives and processes
 - **Patient safety**: medication incident reporting and adverse drug reactions
 - **Product safety**: ensure quality and integrity of products (e.g. cold chain management)
 - **Employee safety**: protecting employees (e.g. hazardous medications policy)
- Risk management and continuous quality improvement

DESIGNATED MANAGER

MANAGING THE PREMISES

Responsible for ensuring that the standards for accreditation and operation of the pharmacy are maintained

- **Community operations advisors** representing the college conduct pharmacy assessments on the operation of the pharmacy
 - Assess the pharmacy's compliance with operational standards and evaluate the procedures in place
 - Identify opportunities for improvement and provide coaching and guidance to the DM on how to meet standards and regulatory expectations to ensure patient safety and quality care
- DM is provided the opportunity to correct issues and address gaps identified through the pharmacy assessment
 - The operations advisor may revisit a pharmacy for reassessment when more support is required for the DM to understand and address outstanding issues where standards are still not fully met
- In circumstances where, despite the operations advisor's support and guidance, a pharmacy continues to fail to meet standards and requirements, the pharmacy may be referred to the accreditation committee

DESIGNATED MANAGER

MANAGING THE EQUIPMENT AND SYSTEMS

The DM is responsible for the overall operation of a pharmacy, including supervision of staff, facilities, equipment and supplies

- Manage the equipment used in the pharmacy with the purpose of protecting against and mitigating risks associated with the delivery of services
 - Systems and the cold chain
 - Systems and inventory management
 - Systems and record keeping
 - Systems and resources
 - Systems and advertising
- Regulations require the provision of the necessary equipment, systems and staffing needed to allow regulated pharmacy professionals practicing in the pharmacy to meet the standards of practice of the profession
- There is an obligation on designated managers, directors and corporations to support the provision of quality services

DESIGNATED MANAGER

MANAGING THE EQUIPMENT AND SYSTEMS

DM is responsible for the overall operation of a pharmacy, including supervision of staff, facilities, equipment and supplies

Managing the cold chain and product safety

- Ensuring the integrity and quality of products sold in the pharmacy

Key areas of focus are:

- Cold chain management → protect products sensitive to temperature differences (e.g. vaccines)
 - Regular monitoring of temperatures and keeping a temperature log
 - Dedicated vaccine fridge that meets public health standards
 - Back-up plan of action in the event of a power outage
 - Poor cold chain management → product loss → profit loss
- Proper storage of medications and products → dry powders away from moisture, reduce exposure to air, heat, etc.
- Proper application of beyond use dates → compounding, certain medications when opened, products that must be dispensed in original packaging

DESIGNATED MANAGER

MANAGING THE EQUIPMENT AND SYSTEMS

DM is responsible for the overall operation of a pharmacy, including supervision of staff, facilities, equipment and supplies

Inventory Management

- The practice of planning, organizing, and controlling inventory to increase profitability
- Goal is to have the lowest carrying cost without compromising supply and demand balance
- Inventory is usually a pharmacy's **LARGEST** and **LEAST LIQUID** asset
- Three common methods of inventory management are the visual method, periodic method, and perpetual method

What, when, how much inventory should be obtained is based on:

Drug usage (based on reports of drug use + time of year)

Storage space available

Backup supply resources

Costs (acquisition, procurement, carrying, stock out)

DESIGNATED MANAGER

INVENTORY MANAGEMENT

Visual

- Manually checking shelves for medications in low quantities
- Inaccurate and time-consuming

Periodic

- Periodic inventory management (e.g., monthly) where a physical count of all stock is done
- Resource intensive and time-consuming

Perpetual

- Using computers/automated devices to provide an ongoing count of stock
- Uses minimum/maximum data where the computer can suggest how much to order
- May use barcoding to expedite re-ordering process (Universal Product Code [UPC] or hospital unit dosing specific codes)
- After drug delivery, the quantities are entered into the computer system (e.g. through barcoding) then placed on the shelves
- Most efficient

DESIGNATED MANAGER

INVENTORY MANAGEMENT

Using generic products

Keeping low stock quantities especially biologics

Reviewing policies for returns

- E.g. credit from supplier if product is close to expiry versus if expired

Managing unclaimed prescriptions

- Calling patients regularly when medications are not picked up

Monitoring for shrinkage

- Implementing cameras to watch for theft

Stocking items on formularies

- Carry 1 therapeutic equivalent to obtain more purchasing power

DESIGNATED MANAGER

INVENTORY MANAGEMENT

$$\text{Inventory Turnover Rate (ITR)} = \frac{\text{COGS}}{\text{Average Inventory}}$$

TURNOVER: Number of times a store's inventory is sold in a year

The industry benchmark ITR for dispensary is 12. Front store turnover is usually lower

- If too **high**: risk items going out of stock, frustrated customers, excessive staffing costs
- If too **low**: risk stock becoming unfit for sale/expired, and money locked in inventory

Increase turnover by:

1. Reducing stock
2. Investing in advertising and promotion
3. Adding products that attract frequent traffic (post office, food/healthcare services, vaccines)
4. Implementing reminder programs to remind patients about due refills

DESIGNATED MANAGER

INVENTORY MANAGEMENT

CASE STUDY

A pharmacy technician is organizing the pharmacy in preparation for a head office visit the following day. She identifies several natural health products in the over-the-counter area that don't have an NPN number on the packaging.

She brings this information to your attention as the manager. What is the **MOST** accurate response you have for her?

- a) An NPN is not required for natural health products
- b) The products can be sold over the counter with no prescription
- c) These products don't meet the regulatory standards and the reason for ordering needs to be investigated
- d) The products are pending acquiring the NPN and the medication is shipped from a small supplier that offers a much bigger discount

DESIGNATED MANAGER

INVENTORY MANAGEMENT

Expired products

- **Suppliers might take back several expired products for credit, however several companies will only give a certain percentage**
 - Suppliers have online information on which products are returnable and which are not
- **Dispensary should be reviewed regularly to identify medications that will be going out of date**
 - Place a sticker or a note on the product to ensure that expired products will never be dispensed
- **All expired products are taken off the shelf and put into a secure area. If they can't be returned, they should be disposed**
 - Front store products can be discounted before expiration date to reduce pharmacy's loss

DESIGNATED MANAGER

INVENTORY MANAGEMENT

Expired Narcotics



All local destructions must be witnessed by a practitioner, pharmacist, a pharmacy intern, or a pharmacy technician

“The method of destruction must alter or denature the controlled substances to such an extent that consumption has been rendered impossible or improbable.”

- E.g. creating a slurry with soap and water

Record Keeping Requirements

The person carrying out the destruction and the witness must print their name, sign and date a witness document containing the following:

- Name, quantity, strength per unit of destroyed substances
- The date and method of destruction
- The unique destruction identification number (for post-consumer returns)
- A statement indicating that the witness attests to witnessing the destruction of all the listed drugs in the application

- ☐ Every transaction concerning unserviceable stock must be recorded and **kept for at least two years** in an auditable format

DESIGNATED MANAGER

INVENTORY MANAGEMENT

CASE STUDY

You are a pharmacy manager at a community pharmacy that is part of a Family Health Team (FHT). A concerned patient comes to you and says "I have a vial containing 8 tablets of hydrocodone at home, I don't want to keep it at home because my grandchildren might be around. I no longer use this medication; can I flush it down the toilet?" How do you respond?

Choose the LEAST accurate answer.

- a) 'No, do not flush it down the toilet, it must be brought back to the pharmacy for proper disposal'
- b) 'Lock it up in your medication cabinet where it is far out of reach from your grandchildren'
- c) 'The pharmacy can offer you a child proof vial container'
- d) 'Yes, this medication can be flushed down the toilet'

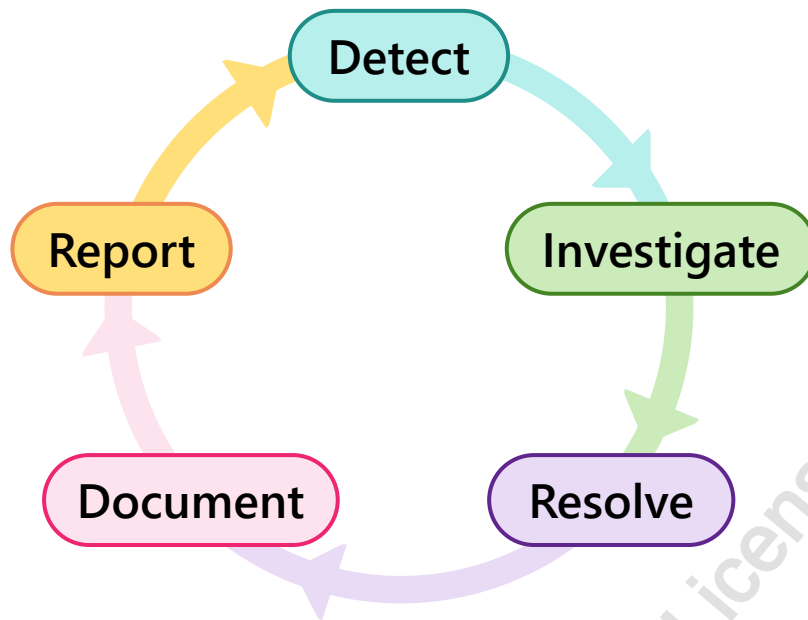
DESIGNATED MANAGER

INVENTORY MANAGEMENT

Reconciliation

The Purpose of inventory reconciliation

To detect, investigate, resolve, document and report losses when there are applicable differences between the amount of inventory you would expect to have on hand and what is actually on hand



- **DETECT** – Identifying discrepancies using reconciliation
- **INVESTIGATE** – Go through records to identify potential discrepancy
- **RESOLVE**
- **DOCUMENT** – Dispensed wrong quantity? Human error? Software error?
- **REPORT** – Unexplained losses, armed robbery, theft, etc. must report to health Canada within 10 days of discovery

DESIGNATED MANAGER

INVENTORY MANAGEMENT

CASE STUDY

You are a pharmacy manager at a community pharmacy that is located within a hospital. Your pharmacy dispenses narcotics and pain medication regularly upon patient discharge.

How often should physical counts and reconciliation of all narcotics, controlled drugs and targeted substances be conducted?

DESIGNATED MANAGER

INVENTORY MANAGEMENT

THEFT CASE STUDY

You are a pharmacy manager at a community pharmacy that is part of a Family Health Team (FHT). It has been brought to your attention that there was a pharmacy robbery of all the oxycodone bottles last night.

Note that the pharmacy does not have a time-delayed safe.

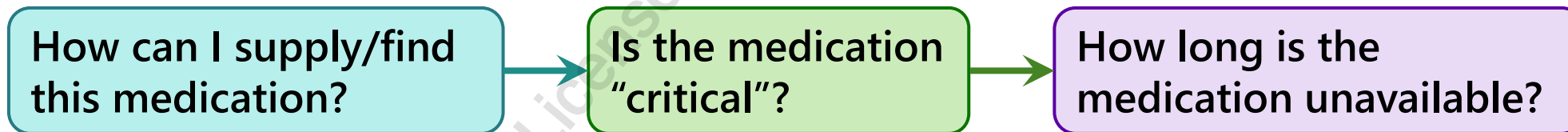
What is the MOST important course of action to take at this time?

DESIGNATED MANAGER

INVENTORY MANAGEMENT

Drug Shortages

- Active pharmaceutical ingredient (API) is now sourced and manufactured around the world with globalization of pharmaceutical manufacturing
- *Factors influencing drug shortages:*
 - Government intervention due to plant inspection or shutdowns
 - Natural disasters
 - Labour shutdowns
 - Unavailability of raw materials
- Tier 3 drug shortage is the most serious with the highest potential impact on drug supply



DESIGNATED MANAGER

INVENTORY MANAGEMENT

Other Causes of Drug Shortages

Supply/demand/sourcing issues

- Discontinuation of a drug (e.g. owing to safety, shift to a new generation of drug, etc.)
- Contamination of active ingredients or raw materials
- Manufacturer poorly estimated demand and can't ramp up production fast enough (e.g. many vaccines have to be cultivated in eggs over many months)
- Sudden demand increase (e.g. demand for Tamiflu® following H1N1 outbreak)

Manufacturing issues

- Voluntary recalls
- Manufacturer has production difficulties (e.g. plant declared unsafe, machine failure, fire/flood)
- Multiple drugs produced using same equipment so increase in production of one (possibly more profitable) drug may mean less production of another (less profitable) drug

Contracting issues

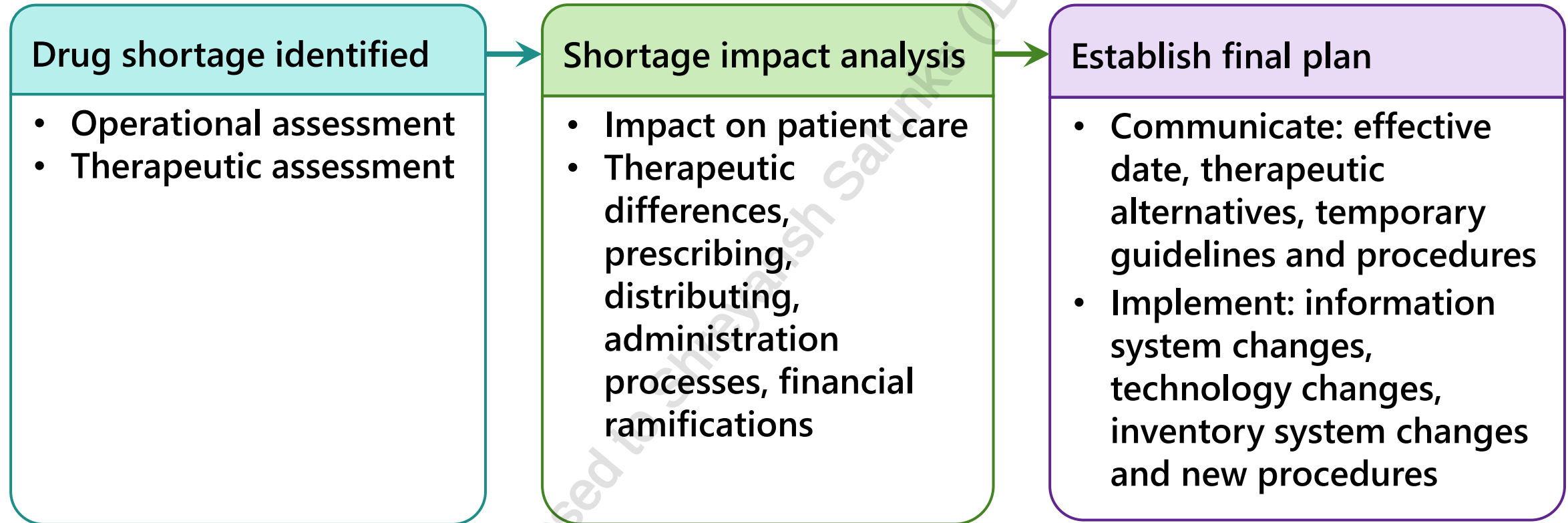
- If you only have one supplier ("sole sourcing") for a product, and that supplier is experiencing a shortage, you may have difficulty accessing product even if other manufacturers still have product available
- Solution: have multiple (or possibly backup) suppliers

Economic issues

- A manufacturer may stop selling a product owing to insufficient profit
- Manufacturers may limit production

DESIGNATED MANAGER

INVENTORY MANAGEMENT



DESIGNATED MANAGER

INVENTORY MANAGEMENT

Drug Shortages: **Must** consider the following questions:

- ✓ Are there any interchangeable brand/generic drug products available?
- ✓ Are there any other dosages or formulations available?
- ✓ Do other pharmacies carry this product?
- ✓ Do other wholesalers carry this product?
- ✓ Can drug supply be obtained directly from the manufacturer?
- ✓ Is the drug/condition critical?
- ✓ What happens if drug is abruptly stopped? (example: beta-blockers, SSRI)
- ✓ Is the medication effective for the condition? Are there any ADR's? Adherence?

DESIGNATED MANAGER

INVENTORY MANAGEMENT

Drug Shortages: Questions to help you navigate

IS THERE AN ALTERNATIVE DRUG?



CAN WE SELECT WITHIN A DRUG CLASS?
Maybe we can consider other drug class? Ex: ACE/ARBs?



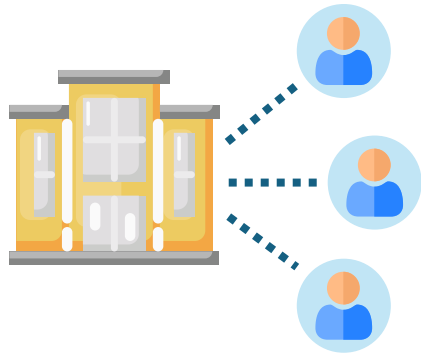
OTHER DRUG CLASSES?			
Is one class of medication better than another? Example: PPI versus H2RA	Has the patient tried another drug class in the past?	ADRs? Contraindications? Allergies? Pregnancy? Interactions?	Cost?

Canadian Pharmacists Association. Drug Shortages: A guide for assessment and patient management. Accessed August 2024.
<https://www.pharmacists.ca/cpha-ca/assets/File/cpha-on-the-issues/DrugShortagesGuide.pdf>

DESIGNATED MANAGER

INVENTORY MANAGEMENT

Drug Shortages: Questions to help you navigate



Collaborate!!!

INITIATING NEW THERAPY



**DO WE DISCONTINUE THE ORIGINAL
DRUG AND START THE NEW ONE?**

Do we need to taper? Overlap? Monitor?



**WHAT HAPPENS WHEN THE DRUG
BECOMES AVAILABLE AGAIN?**

DESIGNATED MANAGER

INVENTORY MANAGEMENT

Drug Shortages: Resources to help you navigate



Drug Shortages Canada | Drug Shortages Homepage



Canadian Pharmacist Association | Drug Shortages

DESIGNATED MANAGER

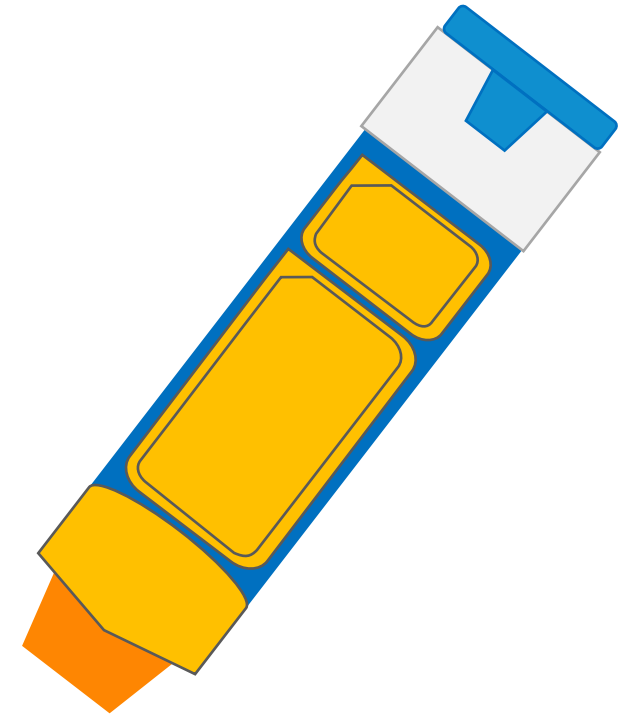
INVENTORY MANAGEMENT

CASE STUDY

Drug Shortages: Questions to help you navigate

WQ is a 25-year-old male who comes to your pharmacy requesting to speak to the pharmacist to get some advice on his EpiPen®. He states he only has one EpiPen® at home that expires next month. He heard from his friend that EpiPens® are currently on back-order and is concerned about potentially needing to use his product at home if he gets an anaphylactic reaction.

How do you respond to WQ's concern?



DESIGNATED MANAGER

INVENTORY MANAGEMENT

CASE STUDY

Drug Shortages: Questions to help you navigate

You are a pharmacist working in a family health team, you receive an email from your colleague mentioning that bisoprolol is on backorder. Many of your patients are currently using bisoprolol to manage their cardiovascular health.

What would be the most appropriate course of action (list the steps)?

DESIGNATED MANAGER

INVENTORY MANAGEMENT

Drug Recalls

Recall



A responsible party's removal from further sale or use, or correction, of a distributed product that presents a risk to the health of consumers or violates the Act or the Regulations.

- Health Canada

- **MedEffect Canada:** Health Canada's advisory and recall program for drugs and other health products
- **Health Canada:** This site supplies public advisories, warnings, information updates, and foreign product alerts

TYPE I

Use of product will cause **serious** health consequences or **death**

TYPE II

Use of product may cause **acute temporary** health effects

TYPE III

Use of product is **unlikely** to cause adverse health consequences

DESIGNATED MANAGER

INVENTORY MANAGEMENT

CASE STUDY

You are a community pharmacist working your regular Monday morning shift. Your assistant hands you a fax that says Pregvit® (pregnancy vitamin) is being recalled. The fax mentions that Pregvit® is a twice-daily vitamin containing a pink vitamin pill in the morning slot and blue vitamin pill in the evening slot. The Pregvit® vitamins with the lot number D0WQ278 are being recalled as some slots have the blue and pink pills mixed.

What type of recall is this?

Do you have to inform all patients about this recall who are using Pregvit®?

DESIGNATED MANAGER

MANAGING THE EQUIPMENT AND SYSTEMS

DM is responsible for the overall operation of a pharmacy, including supervision of staff, facilities, equipment and supplies

Record Keeping and information management

- Designated managers must consider their responsibility in securing all off-site records, including controlling access to areas where records are stored and taking steps to protect records from fire, flooding, viruses and other hazards
- Designated managers must ensure all pharmacists, pharmacy students and interns working at the site are following documentation procedures as per College of Pharmacy guidelines
 - Clinical staff → E.g. Establishing a standard area to document and style to create seamless handover and follow-up, organizational tools to find resources
 - Pharmacists should use SOAP notes and document them in the patient's profile. All pharmacists to check prior to dispensing
 - Pharmacists place prescriptions requiring follow-up in the designated file folder and mark it on the calendar. Upon entering the shift, pharmacists are to check the calendar and address all follow-ups required

DESIGNATED MANAGER

MANAGING THE EQUIPMENT AND SYSTEMS

DM is responsible for the overall operation of a pharmacy, including supervision of staff, facilities, equipment and supplies

Systems and Resources

DM is responsible for adhering to standards regarding information technology and the expectations regarding resources, equipment and technology required for pharmacy staff.

- Record retention and the length of time records must be kept, as well as the use of backup drives or cloud storage
- Encryption to ensure privacy and that appropriate clinical resources and references are available and all pharmacy staff can access those resources
- DMs need to be aware of their responsibility to keep up to date on other requirements, including the use of technology in delivering pharmacy services

DESIGNATED MANAGER

MANAGING THE EQUIPMENT AND SYSTEMS

DM is responsible for the overall operation of a pharmacy, including supervision of staff, facilities, equipment and supplies

Systems and Advertising

A DM or pharmacy owner should consider the following questions:

- What am I hoping to achieve with a social media strategy?
- How am I most comfortable representing myself on social media?
- Who is my target audience, and how are they using social media?
- What is the focus or specialty offered?

Advertisements must abide by provincial and federal legislation

DESIGNATED MANAGER

MANAGING THE PERSONNEL

DM is responsible for the overall operation of a pharmacy, including supervision of staff, facilities, equipment and supplies

- DM has authority and accountability over decisions affecting the operation of the pharmacy
- It is the responsibility of the DM to effectively oversee the day-to-day management of the pharmacy, including all aspects of the pharmacy operation
- DM must ensure that pharmacy staffing supports the pharmacist in their cognitive and patient care functions
- DM must establish job descriptions and performance measurement and improvement system for pharmacy employees
- Every owner or DM of a pharmacy is liable for every offence committed against the Drug and Pharmacies Regulation Act by any person in the employ of, or under the supervision of the owner or designated manager with the owners or designated managers permission, consent or approval express or implied

DESIGNATED MANAGER

MANAGING THE PERSONNEL

Staff Management and assigned roles

Staff types include managers, clinical staff, and support staff

- Managers: oversee operations and administration
- Clinical staff (pharmacists): ensure patient safety and appropriateness of therapy
- Support staff (technicians, assistants, cashiers, etc.): perform technical tasks and provide customer service for patients

Utilizing support staff to their full scope is essential for optimal time management

- Pharmacy technicians managing compliance pack billing and technical checks
- Cross-training staff in all areas for cross-coverage
- Having support staff triage phone calls and patient concerns for clinical staff
- Assigning roles to each member to create consistency
- Ensuring clinical staff develop a process for handover and communication
- Creating a positive work environment through support and team building

DESIGNATED MANAGER

MANAGING THE PERSONNEL

CASE STUDY

Staff Management and assigned roles

The pharmacy you work at received a shipment of influenza vaccinations last week. While you work in the dispensary, the pharmacy owner has been administering vaccinations. You are both tired and overwhelmed as the pharmacy is very busy during flu season. The owner asks if you can work 2 hours more per shift to help him finish up tasks. He is willing to pay you a higher hourly rate for those additional hours.

- Is this an effective way to handle workload?
- Is there a more efficient time management strategy?

DESIGNATED MANAGER

MANAGING THE PERSONNEL

Staff Management and managing tasks

- **Recognize** the need for improvement → identify behaviors requiring change
- **Conduct** an analysis of current time use → track time spent on various tasks
- **Establish mission and set goals** → set short and long term “SMART” goals for improvement
 - **SMART** → Specific, Measurable, Achievable, Realistic, Timely
- **Use organization tools** → use of calendars, folders, binders to reduce wasted time
- **Develop an action plan** → formulate tangible steps to reach set goals
 - E.g., the goal is to reduce time lost searching for patient prescriptions by 25%. Part of the action plan is to develop an efficient organization system sorting by alphabetic order
- **Review, revise, modify** → reflect on the results after applying the action plan, modify and make improvements

DESIGNATED MANAGER

MANAGING THE PERSONNEL

Staff Management and employee safety

Ensuring that adequate interventions and policies are in place to protect employees while working

Key areas of focus are:

- Personal protective and cleaning equipment → hand sanitizers, masks, face shields, gloves, lab coats, etc.
- Policies and procedures → hazardous medications policies, safe compounding procedures, harassment and violence policies
 - E.g. Hazardous and cytotoxic medications should not be modified (e.g. splitting tablets)
- Safe workplace environment → adequate lighting, clean work environment, proper ventilation and sterility
 - E.g. Functioning laminar flow hoods for compounding

DESIGNATED MANAGER

RISK MANAGEMENT

- **Risk management** → Reducing the likelihood of events that can threaten the ability of a person or organization to accomplish its mission
- **The two basic types of risk are speculative and pure risks**
 - Speculative risks → risks with a chance of benefit and a chance of loss (e.g., purchasing stocks)
 - Pure risk → risks with no opportunity for benefit (e.g. natural disasters)
- **Risk management strategies**
 - Risk avoidance → do not pursue high risk activities (e.g. sterile compounding without proper resources)
 - Risk prevention/modification → reduce the risks involved (e.g., developing a process for patient identification to reduce risk of provide medications to the wrong patient)
 - Risk absorption/retention → accepting the risks/losses (e.g., shoplifting, expiring meds)
 - Risk sharing or transfer → share or transfer risk with another party (e.g. purchasing insurance)

DESIGNATED MANAGER

RISK MANAGEMENT

Medication incidents and quality improvement

According to the Institute for Safe Medication Practices in Canada (ISMP), a medication incident is “any preventable event that may cause or lead to inappropriate medication use or patient harm while the medication is in the control of the healthcare professional, patient, or consumer.”

Can originate at any stage of the medication use process:

- Preventable
- Example: incorrect patient profile, incorrect drug, dose, route, frequency entered, etc.

Common factors that increase medication incidents

Staff shortages

Noisy or cluttered workplace setting

Interruptions

Miscommunication

Missing or insufficient documentation

Lack of policy and procedures

Missing patient or drug information

Deficiency in staff or patient education

Medication sound- alike names or look- alike bottles

DESIGNATED MANAGER

RISK MANAGEMENT PROCESS

1. **Identify and analyze risks** → evaluate and prioritize risks → select and implement risk management strategy → monitor and update strategy
 - Example: 3 different pharmacists have dispensed fluoxetine instead of fluvoxamine in the past month
 - Are the bottles held beside each other? Are they properly organized? Do the medications look the same? Are some patients on both medications?
2. **Evaluate and prioritize risks** → Priority is the bottles being placed together and that they look similar
3. **Select and implement risk management strategy** → Risk prevention by placing the bottles on different shelves with dividers, providing education and/or a product identification tool
4. **Monitor and update strategy** → Track how many similar incidents occur monthly for the next 3 months and see if any improvements need to be made

DESIGNATED MANAGER

RISK MANAGEMENT

CASE STUDY

You receive a call at the pharmacy from the owner who sounds upset. He says that a patient came in to fill digoxin 0.0625 mg tablets and was given 0.125 mg tablets during your shift. You review the paperwork attached to this prescription and notice that you did sign off on the prescription. You also remember that two new assistants worked their first shift at the pharmacy during your shift, so it was likely they filled the wrong tablets.

- What do you tell the pharmacy owner?
- Do you tell the owner that it was the assistants' fault?
- What is the most appropriate way to handle this situation?

CONTINUOUS QUALITY IMPROVEMENT (CQI)

- Using a **total systems perspective** on a continuous basis to improve all aspects
- Focuses on the **system as a whole**, instead of the individual
- **Most models include the following core concepts: plan, design, measure, assess, improve**
- Involves an interdisciplinary team to obtain diversity and address gaps from all perspectives
- Steps to performing CQI
 - 1) Determine focus and description → which area are we looking to improve and how do we explain this to the team?
 - 2) Literature review and set goals → how have others handled this situation and what are we aiming to achieve?
 - 3) Implement interventions and measure outcomes → execute the plan and track outcomes
 - 4) Analyze data and develop conclusions, implications and recommendations → what was the end result? What recommendations has the team determined from the CQI? What process changes will occur as a result?

CONTINUOUS QUALITY IMPROVEMENT

EXAMPLE: a hospital has reported 5 fatalities due to improper administration of concentrated electrolytes in the emergency department

Determine focus and description

- Focus is to reduce mortality and medication errors related to concentrated electrolytes
- Discuss with interdisciplinary team to describe the scenario from all perspectives

Literature review and set goals

- Review electrolyte policies of other facilities, research, etc. to determine the intervention
- Set SMART goals such as a 50% reduction in incidents within 6 months

Implement interventions and measure outcomes

- Execute plan to store concentrated electrolytes in a separate area, update electrolyte policies and procedures, educate staff, implement pharmacist intervention such orders
- Develop a tracking system to report the amount of orders, errors, interventions, etc.

Analyze data and develop conclusions, implications and recommendations

- Interdisciplinary teams analyze data and provide reports of the findings
- E.g., after implementing pharmacist interventions over 6 months, there was a 64% reduction in medication errors with concentrated electrolytes concluding that pharmacist intervention played a role, and all concentrated electrolyte orders should involve a pharmacist

ADMINISTRATIVE CONSIDERATIONS

Salary administration	• Ensure pay equity, consistency, salary grade based on performance
Human rights	• Ensure universality in the treatment of staff, ensure freedom of speech, protect against discrimination, harassment and bullying
Liability	• Take responsibility for actions leading to harm and negative consequences
Onboarding	• Structured orientation with on-site training including fire/safety protocols • Scheduling: consistency and equity in shift distribution
Performance appraisals	• Assess performance and areas for improvement • Recognition: formal versus informal, group versus individual
Human resource planning	• Replacement of vacant positions, establish qualifications for newly created positions • Ensure staffing requirements based on workload
Meeting the Employment Standards Act	• Meeting criteria for hours worked, breaks, lunch, vacation time, payment for public holidays, notice for termination, maternity leave benefits, deductions (income tax), pay equity, etc.

COMMON WORKPLACE DEPARTMENTS

Accounting (recording and reporting transactions)	Finance (planning, directing, managing of money)	Human resources	Operations	Marketing
• Maintain accounting records ("bookkeeping") • Record business transactions • Prepare financial statements • Calculate taxes owed • Accounts payable: pay suppliers • Accounts receivable: invoice and collect funds from customers	• Determine financial needs • Source capital (e.g. loans, equity) • Develop budgets • Invest excess funds • Manage assets (e.g., cash, accounts receivable, computers, dispensing equipment) • Procure needed equipment (e.g., lease vs. buy)	• Prepare job descriptions • Hire / fire / retire personnel • Train personnel • Manage personnel • Evaluate personnel • Determine compensation levels • Administer benefits (e.g., health, dental) • Handle workplace complaints	• Determine pharmacy layout • Define jobs and work to be performed • Define processes • Purchase inventory (including drugs, merchandise) and equipment • Complete workflow (e.g. dispense prescriptions, collect funds from customers)	• Assess strengths and weaknesses • Identify competitors • Identify possible target markets • Strategy for 4-P's (product, price, promotion, place) Distribution strategy development

DESIGNATED MANAGER

PRACTICE QUESTIONS

- Q1. You hire a new pharmacy assistant for your community pharmacy. You notice that she has been coming late to work daily for the past week. You also notice she is not abiding by the uniform rules even though she was informed of this information during orientation. What is the MOST appropriate action to take?
- a) Fire her from work
 - b) Explain why what she is doing is wrong in front of other staff, so they understand and do not make the same mistake
 - c) Give a verbal warning
 - d) Suspend her from work
- Q2. You are a manager hiring for a pharmacist position at your community pharmacy. What is the MOST important question to ask during the interview?
- a) Do you have prior pharmacist experience?
 - b) Are you licensed in this province?
 - c) How many hours per week can you work?
 - d) Do you speak other languages?

DESIGNATED MANAGER

PRACTICE QUESTIONS

Q3. AC is a pharmacist who receives the following prescription for David. Upon speaking to the patient, AC deduces that the prescription is for treatment conversion from male to female gender. AC comes to you requiring advice about dispensing these medications since she is not comfortable dispensing them. As the manager, what is the MOST appropriate way to respond to AC?

- a) Remind AC of professional responsibility to fill RX and advise to dispense RX
- b) Remind AC about pharmacist's oath to do no harm and advise to dispense RX
- c) Respect AC's belief system and advise to refer David to other pharmacist or pharmacy
- d) Advise AC to dispense RX at this time and schedule a time for discussion later

RX

Estradiol 1mg one tablet daily

Spironolactone 100mg once daily

Quantity = 30 tabs for both

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CHANGE LOG

February 2025

- Added legend
- Slide 6: Some changes under internal factors
- Slide 15: Rephrased/added info under balance sheet, income sheet and cash flow statement
- Slides 15 & 20: Changed from 'statement of cash flows' to 'cash flow statement'
- Slide 22: Changed equations of gross profit margin and net profit margin
- Slide 23: Changed the question to ask about the gross profit margin of all sales, not just the prescription sales
- Slide 30: Added details for clarification
- Slide 36: Changed ITR equation and changed average turnover of 4-6 to industry benchmark ITR of 12
- Slide 53: Changed "Class" to "Type", minor edits in the definitions of Type II and III drug recalls
- Updated reference list
- Grammatical and formatting edits made throughout

August 2025

- Minor formatting edits made throughout
- Slide 41: added record keeping length
- Slide 50: Updated CPhA's link on drug shortages

October 2025

- Slide 39: Updated record keeping requirements
- Slide 44: Added information on tier 3 drug shortage
- Slide 73: Updated references